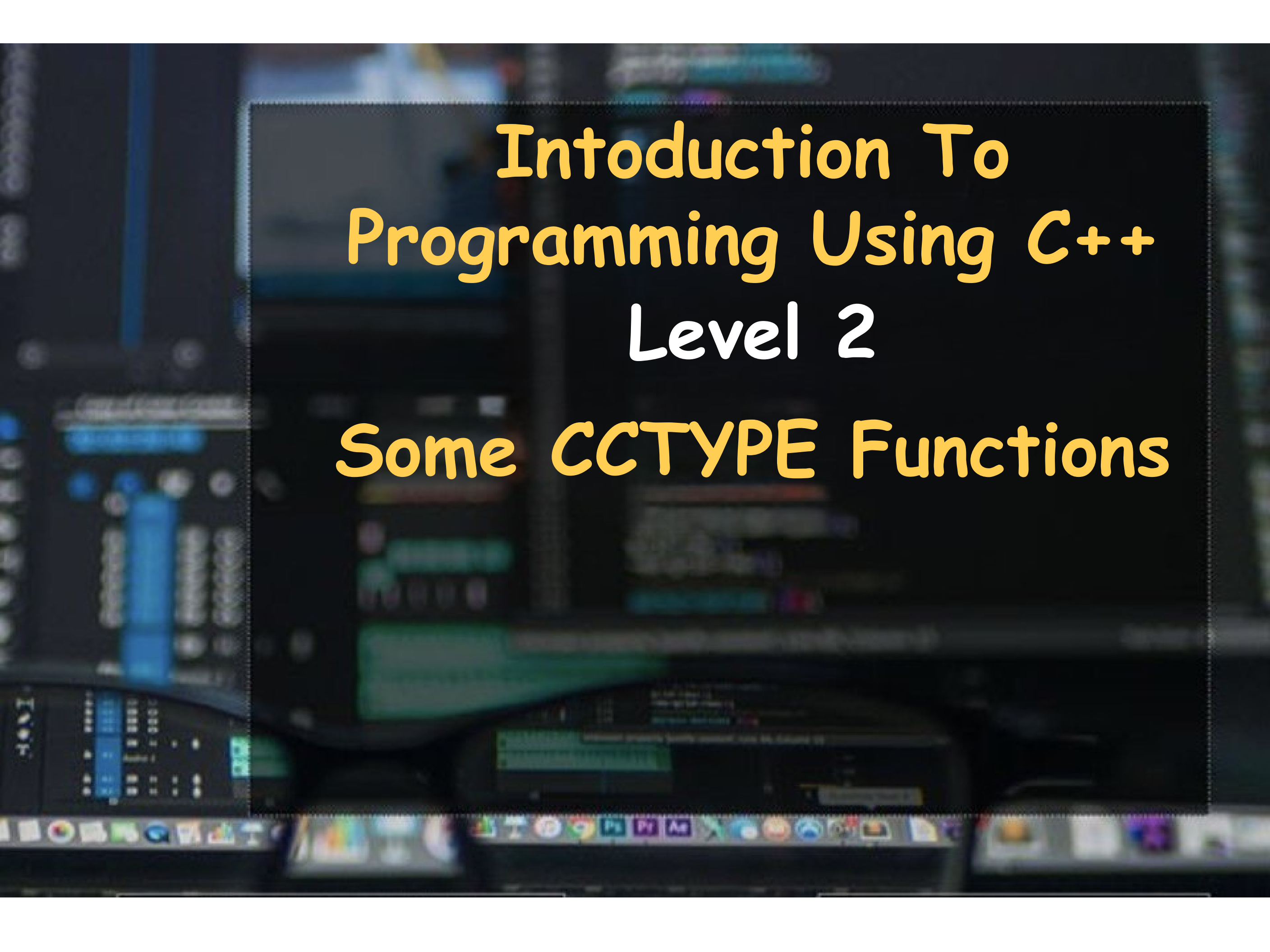
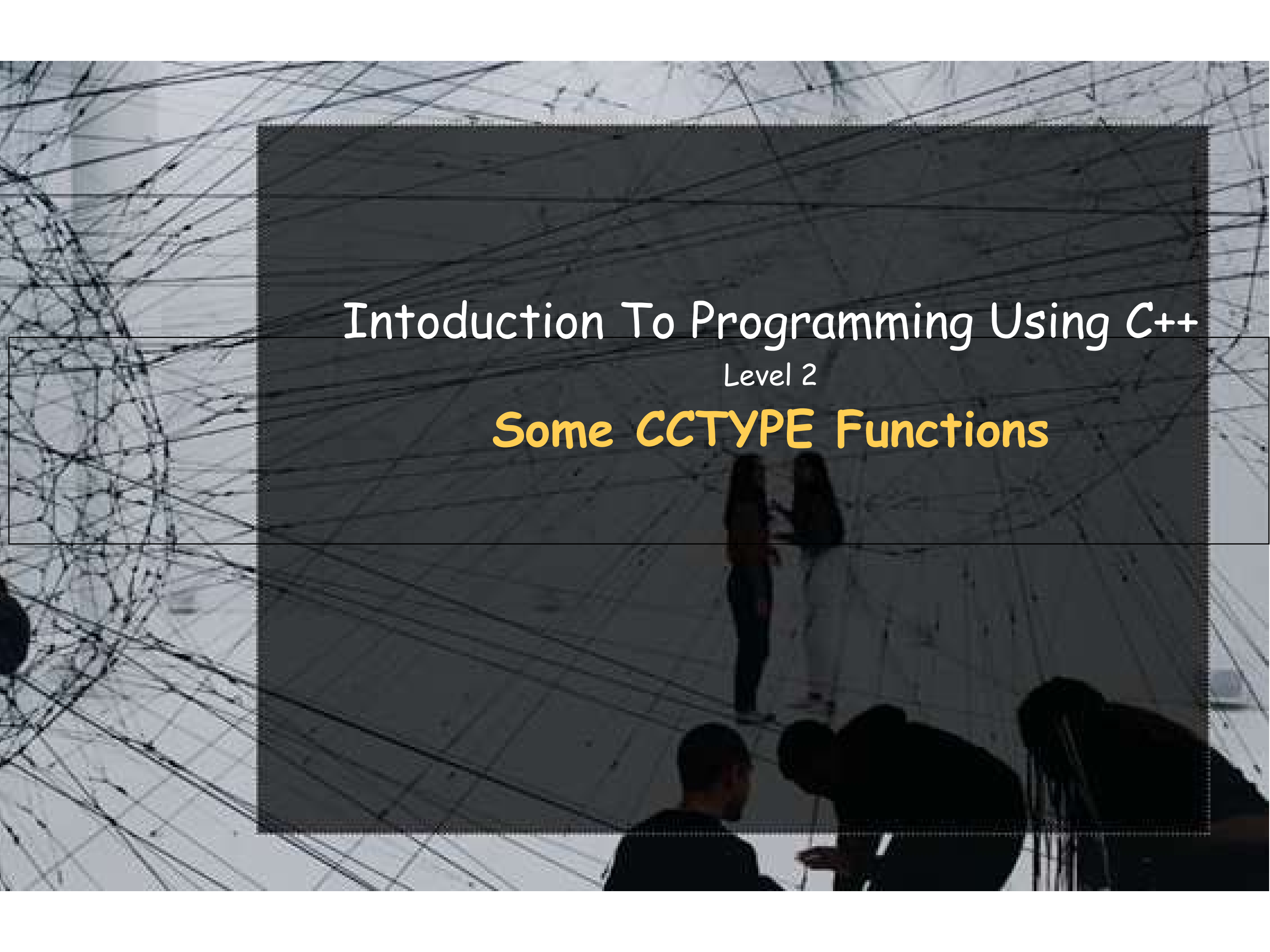




Introduction To Programming Using C++ Level 2

Some CCTYPE Functions



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Some CCTYPE Functions

🔍 Definition

■ ``<cctype>``

▨ The ``<cctype>`` library in C++ provides a set of functions for working with individual characters.

▨ These functions can be used to check the type of a character or convert its case.

▨ Examples include checking whether a character is uppercase, lowercase, a digit, or punctuation.

`#include <cctype>`

Procedures

1. Include the `<ctype>` library.
2. Choose the appropriate character function.
3. Pass a character to the function.
4. For `toupper()` or `tolower()`, use the returned character.
5. For classification functions, use the result in an `if` condition or another expression.
6. Test the function with different characters such as uppercase, lowercase, digits, and punctuation.

💡 Example

```
int main()
{
    char x = toupper('a');
    char w = tolower('A');
    cout << "Converting a to A: " << x << endl;
    cout << "Converting A to a: " << w << endl;
    cout << "isupper('A'): " << isupper('A') << endl;
    cout << "islower('A'): " << islower('A') << endl;
    cout << "isdigit('A'): " << isdigit('A') << endl;
    cout << "ispunct('!'): " << ispunct('!') << endl;
    return 0;
}
```

➡ Result

```
Converting a to A: A
Converting A to a: a
isupper('A'): 1
islower('A'): 0
isdigit('A'): 0
ispunct('!'): 1
```

Note

- ◆ ``toupper()`` and ``tolower()`` are conversion functions.
- ◆ ``isupper()``, ``islower()``, ``isdigit()``, and ``ispunct()`` are character classification functions.
- ◆ Although these functions return an integer type, their result can be used directly in an ``if`` condition.
- ◆ In modern C++, these functions are commonly used with ``char`` values, but technically their parameter type is ``int`` and it should be either ``EOF`` or a value representable as ``unsigned char``.

⊘ Warning

◆ Do not confuse `` functions with functions that work on complete `string` objects.

◆ `` functions examine individual characters, not an entire `string`.

◆ For example:

`isdigit('5');` // ✓ Character

`isdigit("123");` // ✗ Not a character

◆ Be careful when passing a plain `char` containing a negative value to `` functions; for arbitrary byte data, converting to `unsigned char` first avoids undefined behavior.

✦ Summary

- ✓ `<cctype>` provides ready-made functions for working with characters.
- ✓ `toupper()` converts a character to uppercase.
- ✓ `tolower()` converts a character to lowercase.
- ✓ `isupper()` checks for uppercase letters.
- ✓ `islower()` checks for lowercase letters.
- ✓ `isdigit()` checks for digits (`'0-9'`).
- ✓ `ispunct()` checks for punctuation characters.
- ✓ Classification functions return zero for false and a non-zero value for true.
- ✓ `<cctype>` functions are intended for individual characters, not complete strings.

Thank you 😊